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ABSTRACT

This paper concerns the intermittent evolution routes to the asymptotic regimes in the Rulkov map. That is, the windows with transient approximate periodic and transient chaotic behaviors occur alternatively before the system reaches the periodic or the chaotic orbits. Meanwhile, the evolution routes to chaotic orbits can be classified into different types according to the windows before reaching asymptotic chaotic states. In addition, the initial values can be regarded as a key factor affecting the asymptotic behaviors and the evolution routes. The effects of the initial values are given by parameter planes, bifurcation diagrams, and waveforms. In order to investigate whether the intermittent evolution routes can be learned by machine learning, some experiments are given to understanding the differences between the trajectories of the Rulkov map generated by the numerical simulations and predicted by the neural networks. These results show that there is about 60% accuracy rate of successfully predicting both the evolution routes and the asymptotic period-3 orbits using a three-layer feedforward neural network, while the bifurcation diagrams can be reconstructed using reservoir computing except a few parameter conditions.

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Intermittency exhibits transitions in time between periodic and chaotic behaviors. This may occur in different ways, such as type I (saddle-node bifurcation), type II (subcritical Hopf bifurcation), type III (inverse period-doubling bifurcation), on-off intermittency, crisis-induced intermittency, and multistate intermittency. It is observed in a variety of systems in hydrodynamics, thermoacoustics, economics, and neuroscience. Transient behavior in the brain might play an integral role in perceptual phenomena associated with stimuli and in cognitive function.[1,2] In the present paper, we find that there also exist intermittent behaviors, which can be regarded as a kind of evolution route from an initial point to the asymptotic regimes, i.e., the long term behaviors, over time in the chaotic Rulkov neuron model. That is, the windows with transient approximate periodic and transient chaotic behaviors occur alternatively before the system reaches the period-k ($k \geq 3$) or the chaotic orbits. Due to the characteristics of the routes, the evolution routes to chaos can be sorted as different types according to the windows before reaching asymptotic chaotic states. In addition, the initial values can be regarded as a key factor affecting both the asymptotic behaviors and the evolution routes in the Rulkov map. Based on the complexity of the intermittency evolution routes, we also focus on whether the routes can be predicted by using feedforward neural networks (FFNNs) and discuss the performances of reservoir computing when reconstructing the bifurcation diagrams of a single Rulkov model and the synchronous Rulkov network model.

I. INTRODUCTION

Rulkov map, which is always called the chaotic Rulkov model, is an important two-dimensional neuron model in computational neuroscience.[3–6] It is given by the map F,

$$\begin{cases} x_{n+1} = \frac{\alpha}{1+x_n^2} + y_n, \\ y_{n+1} = y_n - \eta(x_n - \sigma), \end{cases} \tag{1}$$

where n ($n = 0, 1, 2, \ldots$) is the discrete time, the fast variable x_n represents the transmembrane voltage of a single neuron, the slow variable y_n represents the gating process, x_0 and y_0 are the initial values of the Rulkov map F, and α and σ are control parameters. The parameter η is a sufficiently small parameter. Herein, η is fixed at 0.001.

Rulkov map presents rich periodic and chaotic orbits caused by period-adding bifurcation.[7–9] Figure 1(a) shows the bifurcation diagram with $\alpha = -15$ and varying parameter σ. 400 parameter

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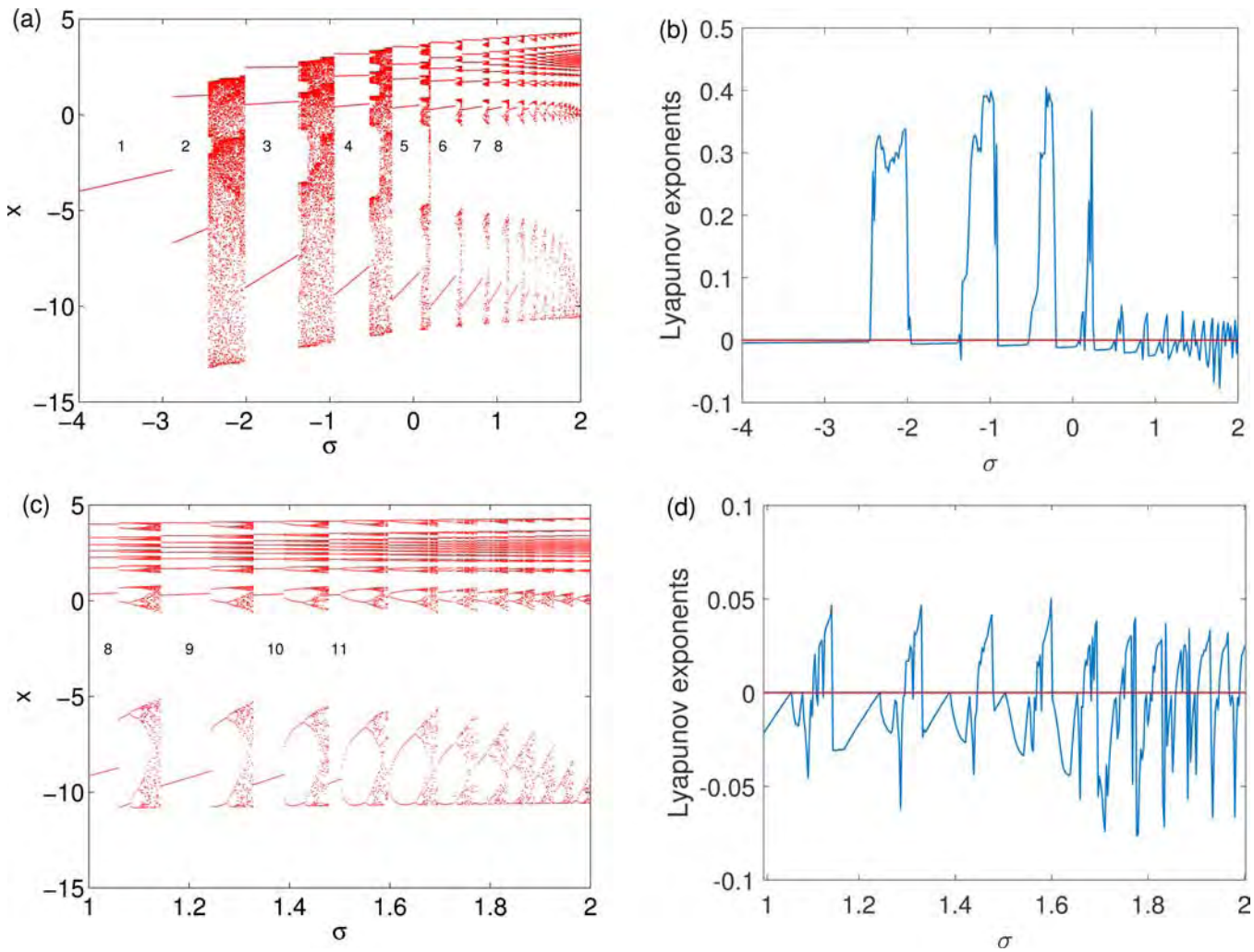

FIG. 1. (a) Period-adding bifurcation diagram about parameter σ when the control parameter $\alpha = -15$. (b) The maximal Lyapunov exponents corresponding to Fig. 1(a). (c) The enlargement version of (a). (d) The maximal Lyapunov exponents corresponding to Fig. 1(c).

values of σ are chosen from the interval $[-4, 2]$ and the map is iterated 50 150 times using the initial values $x_0 = -1$ and $y_0 = -2.7$. The figure presents the last 150 iterations while dropping transient behaviors, i.e., we only consider the periodic or the chaotic orbit after the first 50 000 iterations in this paper. One can see that the asymptotic dynamical behaviors converge to a fixed point $\rightarrow$ period-2 orbit $\rightarrow$ chaos $\rightarrow$ period-3 orbit $\rightarrow$ chaos $\rightarrow$ period-4 orbit $\rightarrow$ chaos $\rightarrow$ period-5 orbit $\rightarrow$ chaos $\rightarrow \cdots \rightarrow$ period-12 orbit $\rightarrow$ chaos $\rightarrow \cdots$ as the parameter σ increases. Corresponding to the parameter values, the maximal Lyapunov exponents are computed and shown in Figs. 1(b) and 1(d).

As shown in Fig. 1(c), some of the period-k orbits become chaotic orbits through a cascade of period-doubling bifurcations as the parameter σ increases. The period-doubling route discovered by Feigenbaum is one of the classic routes to chaos.[10] The other two classic routes to chaos include the Ruelle–Takens–Newhouse (RTN) route and the intermittency route.[11,12] The RTN route involves the birth of a quasi-periodic attractor (3-torus) via three successive Hopf bifurcations. In the intermittency route, the system's oscillating behavior switches between regular and chaotic motions. As parameters keep changing, the irregular motion takes longer and longer time, and finally the system completely enters the irregular motion.[12,13] Pomeau and Manneville identified three types of intermittency routes to chaos: type I (saddle-node bifurcation), type II (subcritical Hopf bifurcation), and type III (inverse period-doubling bifurcation). A few more routes are discovered later, such as on-off intermittency, crisis-induced intermittency, and multistate intermittency.[14–17]

Intermittency exhibits transitions in time between periodic and chaotic behaviors. It is observed in a variety of systems in

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hydrodynamics, thermoacoustics, economics, and neuroscience.[15,18–21] For example, the stable periodic motion of the Lorenz model is interrupted by bursts, which occur frequently as parameter r increases.[12] In a dynamic system with multiple Lorenz attractors, there is a transition layer with regular oscillation between two local chaotic attractors.[22] In a network coupled by Henon maps, a time series is characterized by irregular switching in time between period-8 and chaotic oscillatory modes.[23] The appearance of chaos with finite lifetime is known as transient chaos, which often suddenly settles down to an asymptotic state, i.e., long term behavior differing from transients.[24] In this paper, we find that there also exist intermittent behaviors in the Rulkov map, which can be regarded as a kind of evolution route from an initial point to a period-k ($k \geq 3$) or a chaotic orbit over time. In addition, we will examine whether the artificial neural network can learn the intermittent evolution patterns in the Rulkov neuron model. Some experiments are given in order to understand the performances of a three-layer feedforward neural network and reservoir computing when reconstructing the intermittent evolution routes and the bifurcations of the Rulkov map.

This paper is arranged as follows. In Sec. II, we present the intermittent evolution routes to the periodic or the chaotic orbits in the Rulkov map, which contain the windows with transient approximate periodic behaviors and transient chaotic behaviors. Based on these results, the evolution routes to chaos can be classified into different types according to the windows before reaching asymptotic chaotic regimes. In the Rulkov map, the initial condition is a key factor affecting the asymptotic behaviors and the evolution routes. Therefore, Sec. III studies the impacts of the initial values on the map's behaviors. In Sec. IV, the evolution routes are predicted using the feedforward neural networks (FFNNs). Because of the weakness of FFNNs, we then reconstruct the evolution routes, bifurcation diagrams, and synchronous network using reservoir computing. A conclusion section comes at the end.

II. INTERMITTENT EVOLUTION ROUTES

Figure 1 shows the asymptotic regimes about the parameter regions of the Rulkov map (1). In this section, all the numerical simulations have the same initial point $x_0 = -1$ and $y_0 = -2.7$ as that in Fig. 1. Based on the waveforms and continuous wavelet transforms (CWTs), we explore the evolution routes from the initial point to the asymptotic periodic and the asymptotic chaotic orbits, respectively.

A. Evolution routes to periodic orbits

Figure 2(a) shows the evolution route from the initial values to a stable fixed point $z = (\sigma, \sigma - \frac{\alpha}{1+\sigma^2})$ when $\alpha = -15$ and $\sigma = -3.5$. The phase point of x_n jumps around the fixed point and converges to the fixed point quickly over time. The reason can be given as the following: the Jacobian matrix of the Rulkov map at the fixed point is

$$DF_z = \begin{pmatrix} -\frac{2\alpha\sigma}{(1+\sigma^2)^2} & 1 \\ -\eta & 1 \end{pmatrix}. \tag{2}$$

When $\alpha = -15$ and $\sigma = -3.5$, two eigenvalues of DF_z are $\lambda_1 = -0.5975$ and $\lambda_2 = 0.9994$ corresponding to the eigenvectors

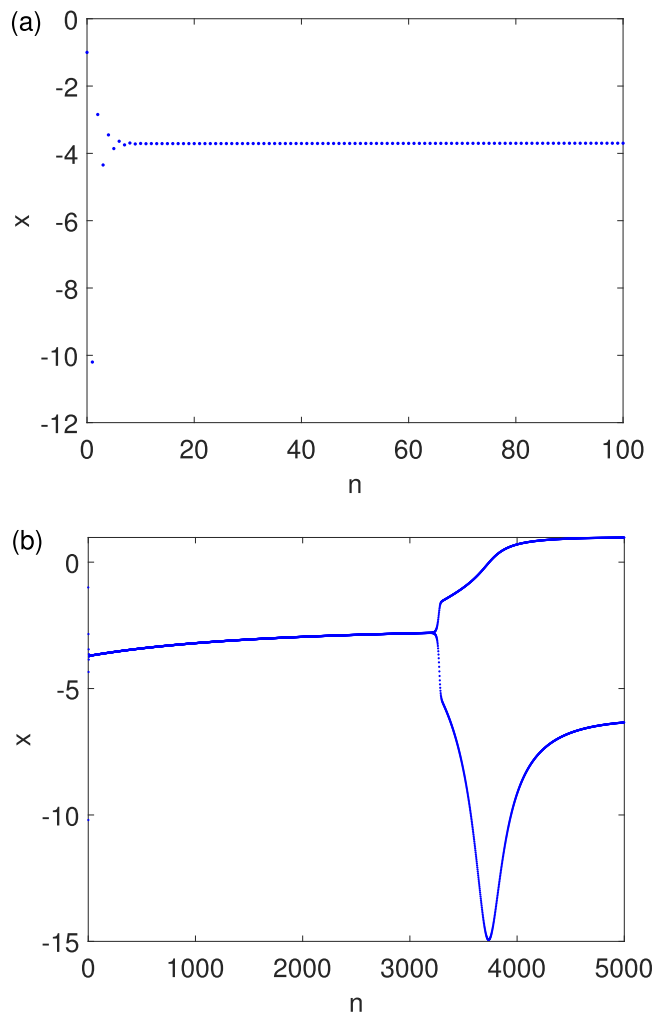

FIG. 2. (a) The evolution routes from the initial values $x_0 = -1$ and $y_0 = -2.7$ to a stable fixed point when $\sigma = -3.5$. (b) The evolution from the same initial values to a period-2 orbit when $\sigma = -2.6$.

$v_1 = [-1.0000, -0.0006]^T$ and $v_2 = [-0.5306, -0.8476]^T$, respectively. z is a stable fixed point since $0 < |\lambda_1|, |\lambda_2| < 1$. "Jumping" orbits can only be observed in discrete-time systems with negative eigenvalue(s), differing from the discrete-time systems with only positive eigenvalue(s) or the continuous-time systems.[25–27]

Figure 2(b) shows the evolution route from the same initial values to a period-2 orbit when $\alpha = -15$ and $\sigma = -2.6$. At the beginning of the iterations, the phase point of x_n also jumps and then converges to a smooth curve quickly. It is similar to that in Fig. 2(a). After about 3000 iterations, the phase point of x_n suddenly separates into two branches and eventually tends to a period-2 orbit.

What would happen if the period is larger than 2? Figures 3(a)–3(i) show the evolution routes from the initial point to different period-k ($k = 3, 4, \ldots, 11$) orbits for the Rulkov map

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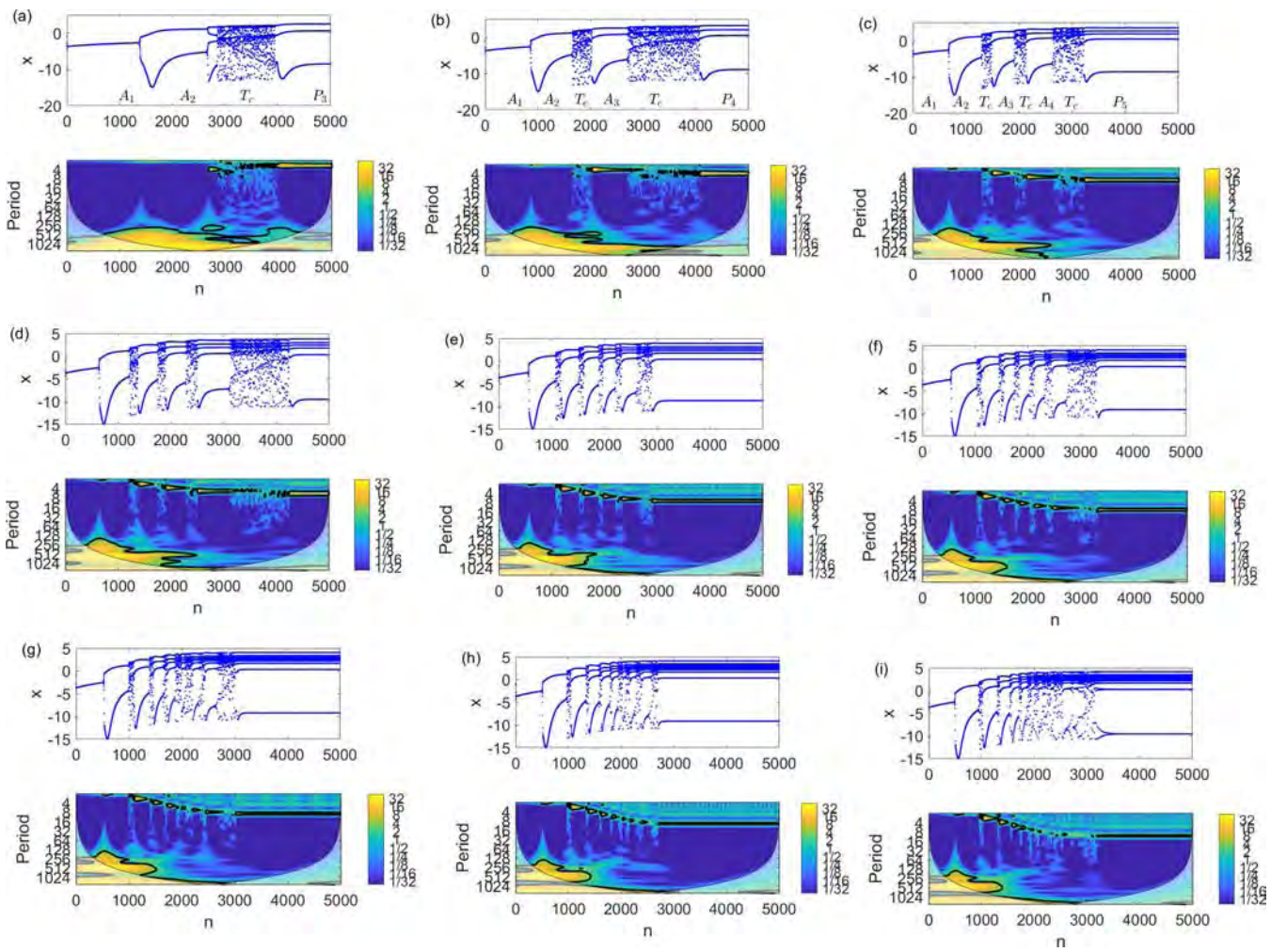

FIG. 3. Evolution routes and their continuous wavelet transforms (CWTs) from the initial point $x_0 = -1$ and $y_0 = -2.7$ to (a) period-3 orbit when $\sigma = -1.8$; (b) period-4 orbit when $\sigma = -0.8$; (c) period-5 orbit when $\sigma = 0$; (d) period-6 orbit when $\sigma = 0.3$; (e) period-7 orbit when $\sigma = 0.8$; (f) period-8 orbit when $\sigma = 1.0$; (g) period-9 orbit when $\sigma = 1.2$; (h) period-10 orbit when $\sigma = 1.38$; and (i) period-11 orbit when $\sigma = 1.48$.

when $\sigma = -1.8$, -0.8, 0, 0.3, 0.8, 1.0, 1.2, 1.38, and 1.48, respectively. Figure 3 also presents continuous wavelet transform (CWT) corresponding to the waveform in the figure. CWT expands time series into time–frequency space and can find localized intermittent periodicities, i.e., the evolution of the frequencies in a time series.[28,29] In the figures of CWTs, the points in the lighter shade region are not reliable due to the characteristics of the method, so they can be omitted. The points in the thick black contour are not completely localized in time because of the limitations of the method.

As shown in Figs. 3(a)–3(f), one can see intermittent windows of transient approximate periodic and transient chaotic behaviors, which occur alternatively before the system reaches asymptotic periodic orbits. For a window with transient approximate periodic behavior, there will be a small yellow area at the corresponding ordinate $k \in Z^+$ in the figures of CWTs. To characterize the evolution routes to the periodic or the chaotic orbits of the Rulkov map, we will call the windows with transient approximate period-k behavior A_k, where k stands for the number of periods. For a window with transient chaotic behavior, there will be multiple light blue areas in the figures of CWTs, which determine that the windows contain chaotic motions rather than periodic oscillations with high periods. A window with transient chaotic behavior will be denoted by T_c. We also denote the asymptotic period-k regime by P_k. The differences between P_k and A_k are: (1) Period-k describes asymptotic behaviors of the system, while A_k happens at a window in the evolution patterns. (2) Period-k makes the time series $x_{n+k} = x_n$, while A_k makes $x_{n+k} \approx x_n$.

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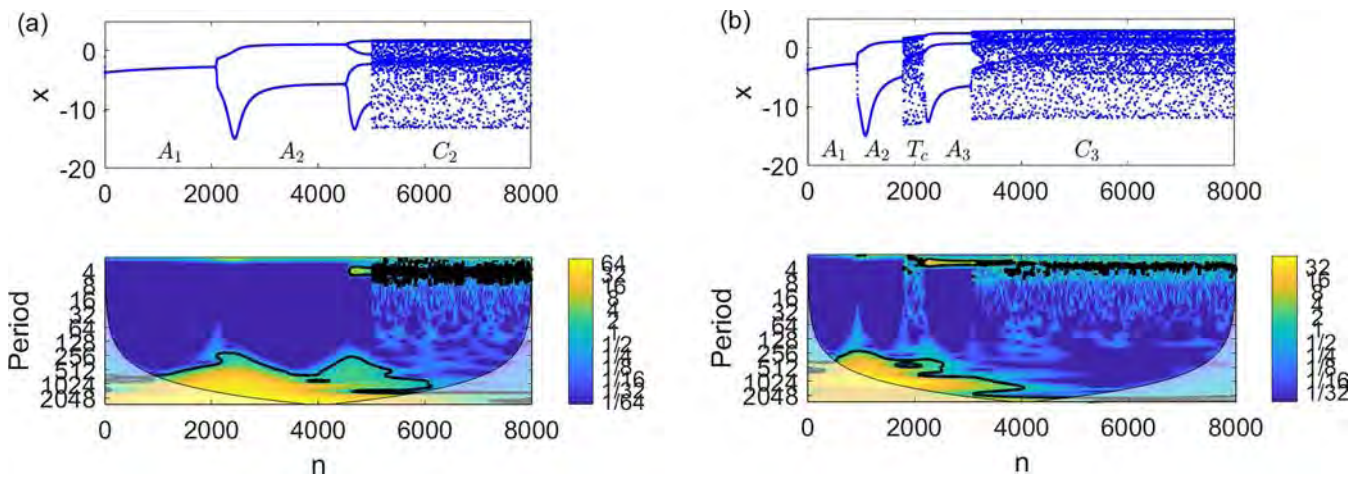

FIG. 4. Evolution routes and their continuous wavelet transforms (CWTs) from the initial point to (a) chaos-2 orbit when $\sigma = -2.3$ and (b) chaos-3 orbit when $\sigma = -1$, respectively.

We have the following intermittent evolution routes to the asymptotic periodic orbits of the Rulkov map based on Fig. 3:

Route 1. There exists an intermittent evolution route from an initial point to a period-k ($k \geq 3$) orbit, that is, *an initial point* $\xrightarrow{converge} A_1 \to A_2 \to T_c \to A_3 \to T_c \to \cdots \to T_c \to A_{k-1} \to T_c \to P_k$.

B. Evolution routes to chaotic orbits

Similar to the periodic orbits, Figs. 4 and 5 show the evolution routes and their continuous wavelet transforms from the same initial point to chaotic orbits over time.

The evolution route in Fig. 4(a) is *an initial point* $\xrightarrow{converge} A_1 \to A_2 \to A_4 \to$ *chaos*, while the route in Fig. 4(b) is *an initial point*

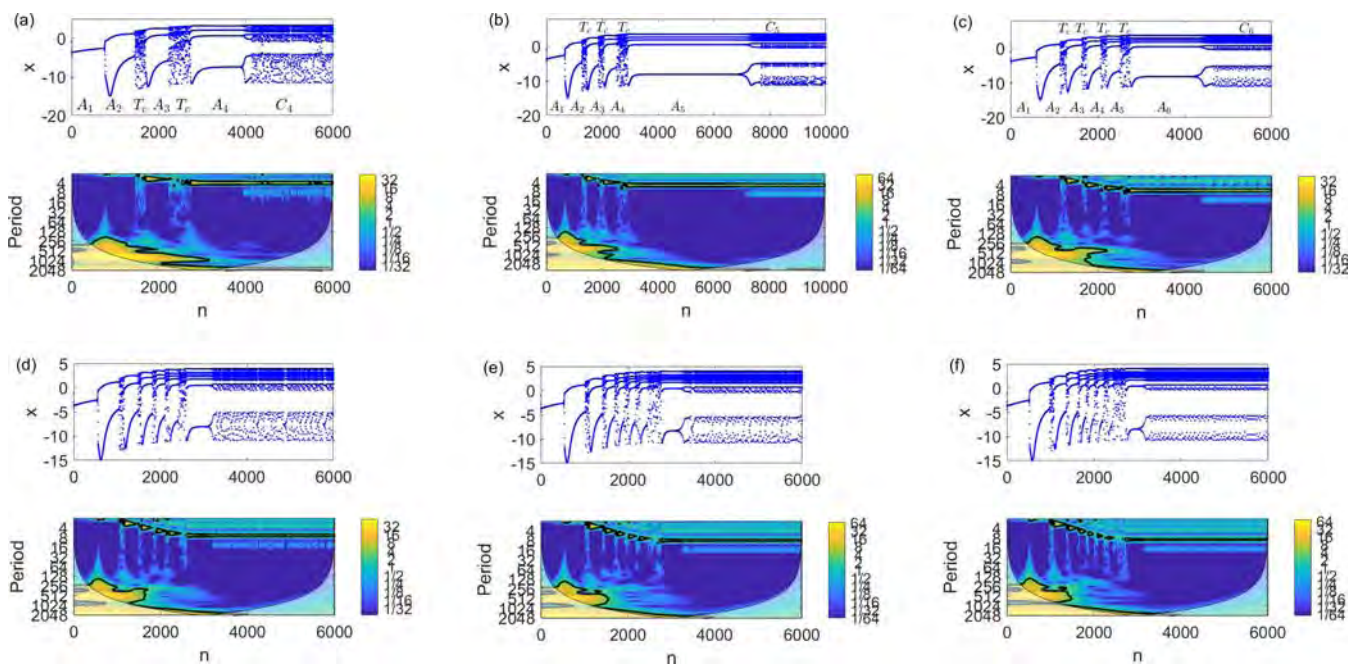

FIG. 5. Evolution routes and their continuous wavelet transforms (CWTs) from the initial point to (a) chaos-4 when $\sigma = -0.4$; (b) chaos-5 when $\sigma = 0.1$; (c) chaos-6 when $\sigma = 0.55$; (d) chaos-7 when $\sigma = 0.9$; (e) chaos-8 when $\sigma = 1.12$; and (f) chaos-9 when $\sigma = 1.3$, respectively.

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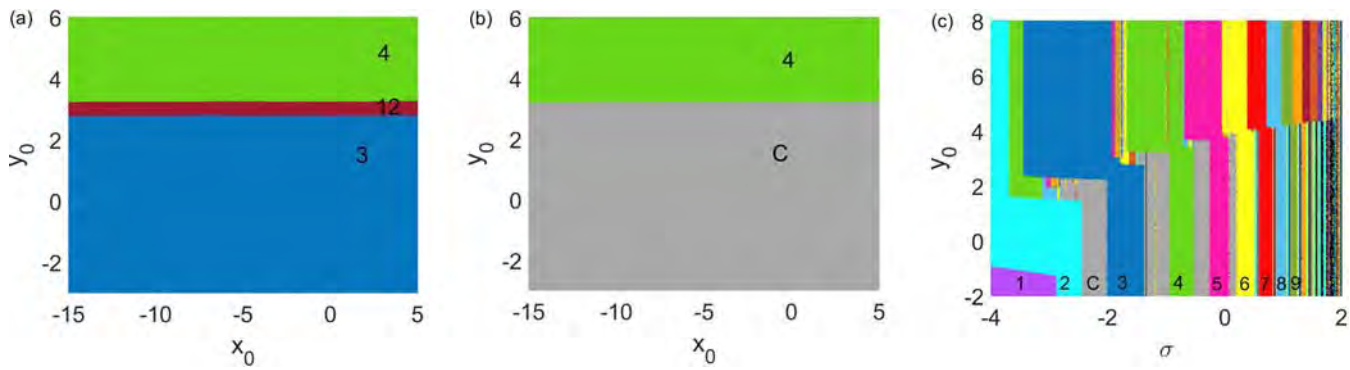

FIG. 6. Asymptotic behaviors of the Rulkov map (1) with different initial values and parameters: (a) the initial values $(x_0, y_0) \in [-15, 5] \times [-3, 6]$ and $\sigma = -1.6$; (b) the initial values $(x_0, y_0) \in [-15, 5] \times [-3, 6]$ and $\sigma = -1.3$; and (c) $(\sigma, y_0) \in [-4, 2] \times [-2, 8]$ and $x_0 = -1$.

$\xrightarrow{converge} A_1 \to A_2 \to T_c \to A_3 \to$ *chaos*. Figures 5(a)–5(f) also show the evolution routes from the initial point to chaotic orbits when $\sigma = -0.4$, 0.1, 0.55, 0.9, 1.12, and 1.3, respectively. The evolution routes to chaos can be summarized as:

Route 2. There exists an intermittent evolution route from an initial point to a chaotic orbit, that is, *an initial point* $\xrightarrow{converge} A_1 \to A_2 \to T_c \to A_3 \to T_c \to \cdots \to T_c \to A_{k-1} \to T_c \to A_k \to$ *chaos*.

Sometimes, there is an A_{2k} between A_k and the asymptotic chaotic orbit in the route, i.e., $\cdots \to A_k \to A_{2k} \to$ *chaos*, like the route in Fig. 4(a). However, there is no obvious A_6 between A_3 and the asymptotic chaotic orbit in Fig. 4(b). Therefore, we do not conclude it as a general result in Route 2. Furthermore, the evolution routes to chaos can be classified into different types according to the windows before reaching asymptotic chaotic regimes. If the window in an evolution route before asymptotic chaotic regime is A_k, we denote the asymptotic chaotic regime by chaos-k (or $C_k, k \geq 2$). In other words, C_k indicates that the evolution route to the asymptotic chaotic orbit is *an initial point* $\to A_1 \to A_2 \to T_c \to A_3$

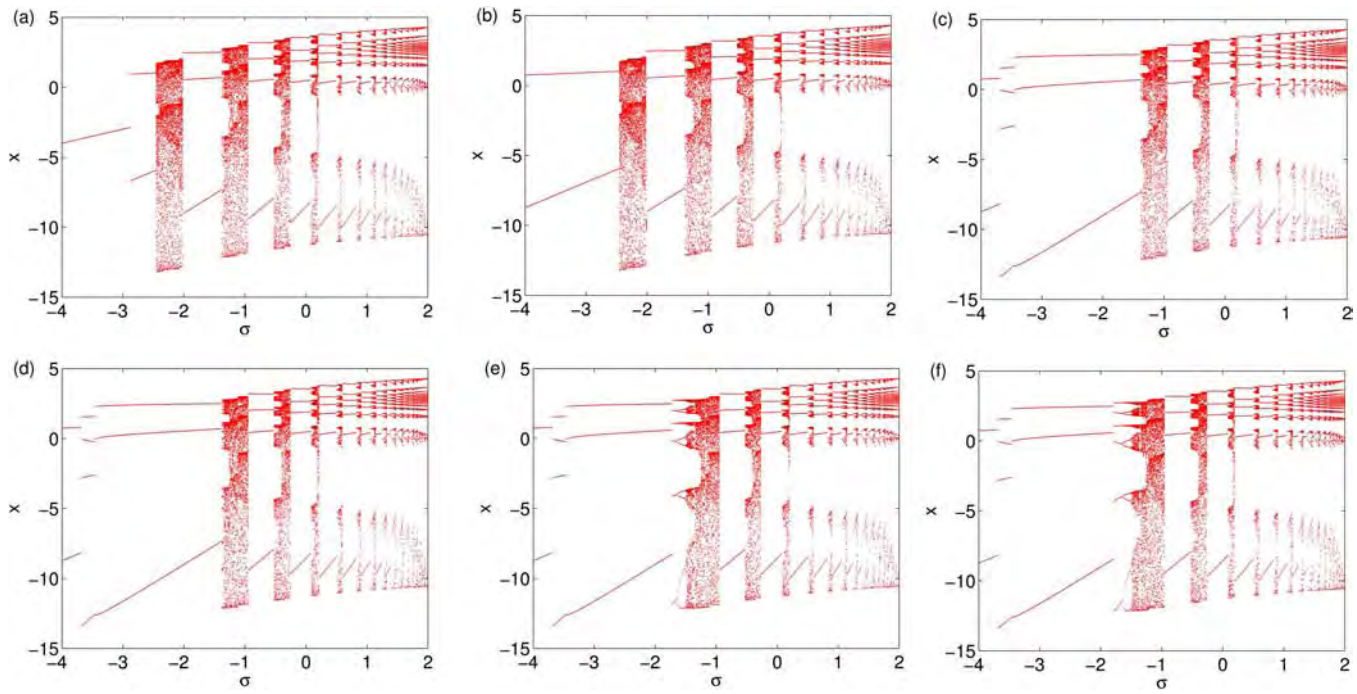

FIG. 7. Bifurcation diagrams about the parameter σ when the initial values are $x_0 = -1$, (a) $y_0 = -2.7$; (b) $y_0 = 1.2$; (c) $y_0 = 2.7$; (d) $y_0 = 2.75$; (e) $y_0 = 2.8$; and (f) $y_0 = 3.0$.

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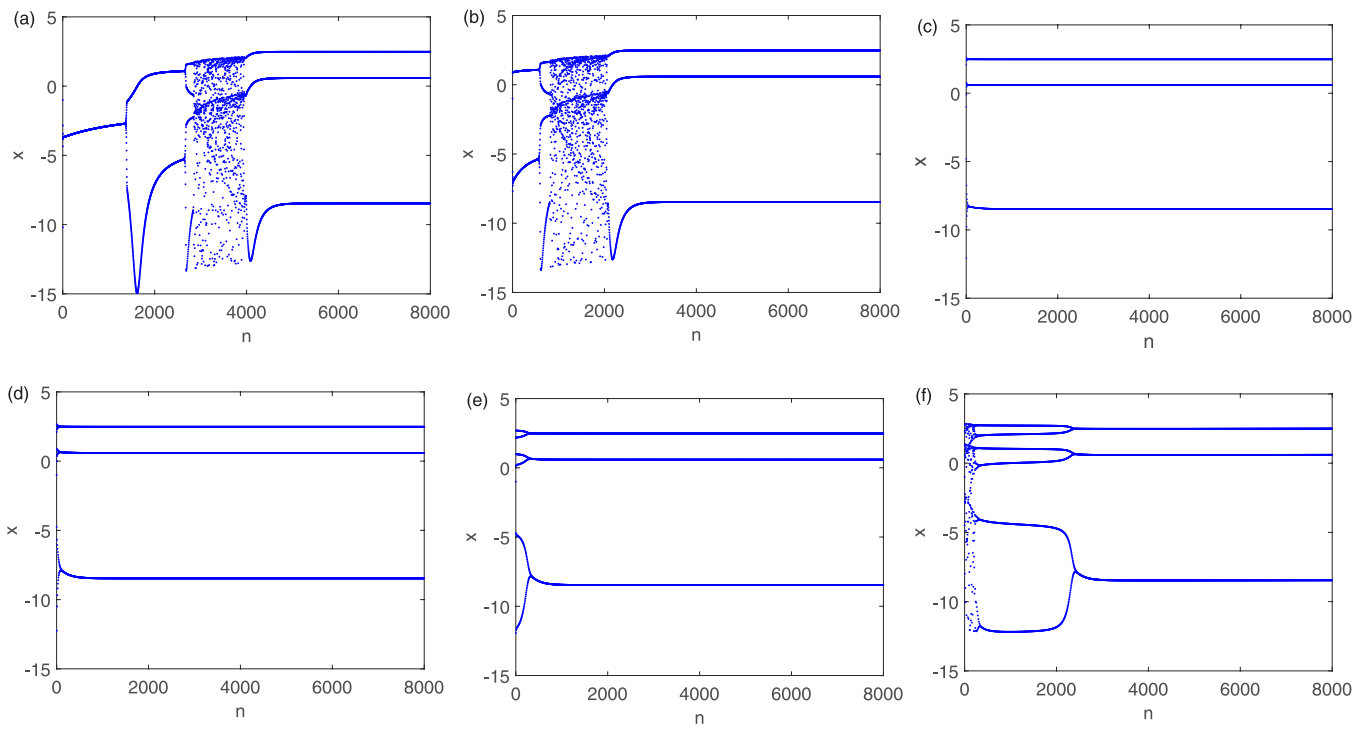

FIG. 8. Evolution routes to period-3 orbit when $\sigma = -1.8$ and the initial point $x_0 = -1$ (a) $y_0 = -2.7$; (b) $y_0 = 1.2$; (c) $y_0 = 2.7$; (d) $y_0 = 2.75$; (e) $y_0 = 2.8$; and (f) $y_0 = 3.0$.

$\rightarrow T_c \rightarrow \cdots \rightarrow T_c \rightarrow A_{k-1} \rightarrow T_c \rightarrow A_k \rightarrow C_k$. For example, the asymptotic chaotic states in Figs. 5(a)–5(f) can be denoted as C_k ($k = 4, 5, \ldots, 9$), respectively. We also find that the evolution route to C_k happens when the parameter σ is chosen between the region of period-k and the region of period-$(k+1)$ in the bifurcation diagrams [Figs. 1(a) and 1(c)].

III. INITIAL VALUES AFFECT BIFURCATIONS AND ROUTES

Route 1 and route 2 are summarized when the initial values are fixed at $x_0 = -1$ and $y_0 = -2.7$ in Sec. II. However, chaos displays sensitive dependence on initial conditions. In this section, we present some different evolution routes to the periodic or the chaotic orbits in the Rulkov map, which can be considered as parts of route 1 or route 2.

A. Initial values affect asymptotic behaviors

Figures 6(a) and 6(b) are two planes about the initial values x_0 and y_0 when $\sigma = -1.6$ and -1.3, respectively. Discretize the region $(x_0, y_0) \in [-15, 5] \times [-3, 6]$ in a grid of 400×400 values. The dynamics of the Rulkov map are displayed in different colors. Figures 6(a) and 6(b) show that the initial value of the slow variable y_0 affects the asymptotic behaviors, while the initial value of the fast variable x_0 has little effect on the asymptotic behaviors. Figure 6(c) presents a plane about the parameter σ and the initial value y_0 when $\alpha = -15$ and $x_0 = -1$. The increase of y_0 will move the parameter plane to the left. Meanwhile, some new periodic regions will appear as the movement happens.

To illustrate the effects on the bifurcations/asymptotic behaviors (either periodic or chaotic), we choose six different values as the initial conditions of the slow variable y_0 in Fig. 6(c). Figures 7(a)–7(f) are six bifurcation diagrams about the parameter σ when the initial values are $x_0 = -1$, $y_0 = -2.7$, 1.2, 2.7, 2.75, 2.8, and 3.0, respectively. The changes of dynamical phenomena are mainly in the parameter range $\sigma < -1.4$: (1) Figure 7(a) is the same figure as Fig. 1(a) when the initial values are $x_0 = -1$ and $y_0 = -2.7$. (2) Figure 7(b) is a bifurcation diagram with the initial values $x_0 = -1$ and $y_0 = 1.2$. The difference between Figs. 7(a) and 7(b) is that the orbits of period-1 become period-2 when $-4 < \sigma < -2.9$. (3) The bifurcation diagram can change little if the initial value of y_0 changes little. This is demonstrated in Figs. 7(c) and 7(d) when $y_0 = 2.7$ and $y_0 = 2.75$, respectively. The *period-2 and chaotic region* at $\sigma < -2$ becomes a *period-2* $\rightarrow$ *period-4* $\rightarrow$ *period-3 region*. (4) Comparing with Figs. 7(c) and 7(d), the *period-3 region* at $-1.9 < \sigma < -1.4$ becomes period-doubling bifurcation region that presents *period-3* $\rightarrow$ period-6 $\rightarrow \cdots$ *chaos* in Figs. 7(e) and 7(f) when $y_0 = 2.8$ and $y_0 = 3.0$, respectively.

Choosing other initial values will lead to more changes in the bifurcation diagrams but only six examples are given here. Although the period-k ($k \geq 4$) and the chaotic regions in Figs. 7(a)–7(f)

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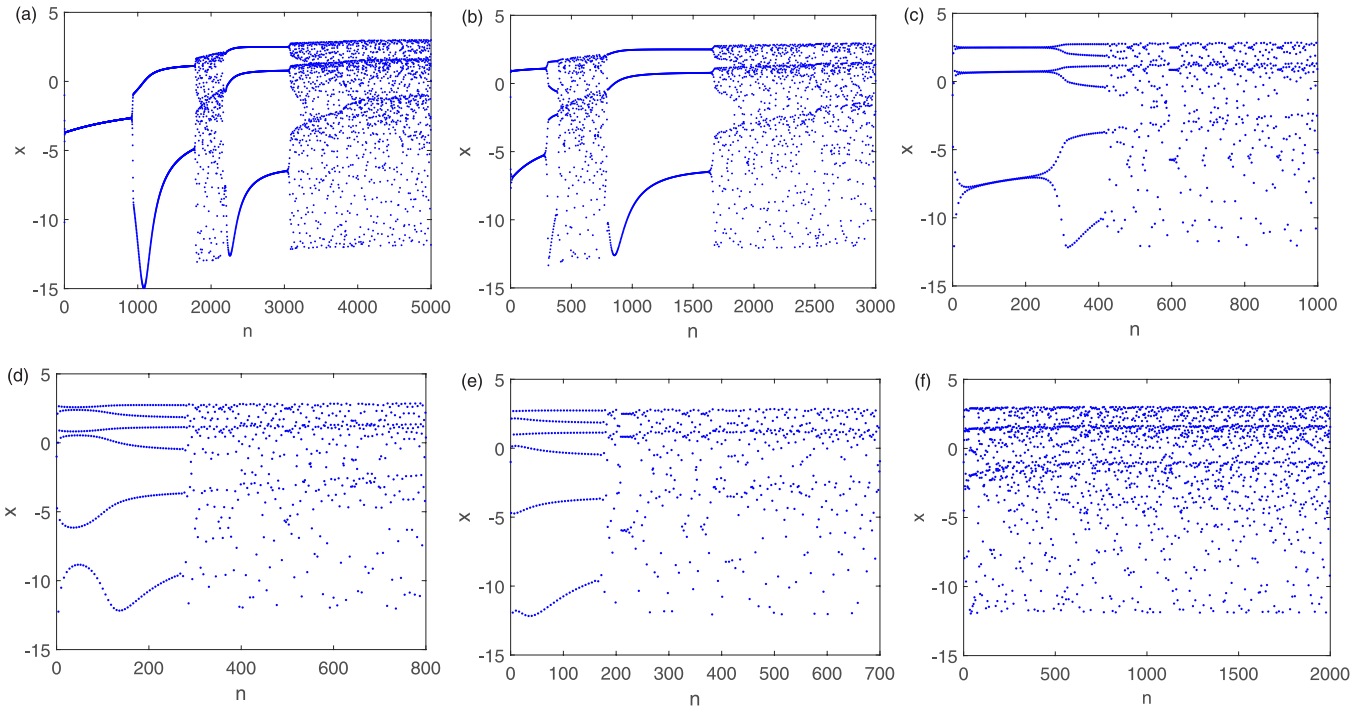

FIG. 9. Evolution routes to chaos-3 orbit when $\sigma = -1$ and the initial point $x_0 = -1$ (a) $y_0 = -2.7$; (b) $y_0 = 1.2$; (c) $y_0 = 2.7$; (d) $y_0 = 2.75$; (e) $y_0 = 2.8$; and (f) $y_0 = 3.0$.

are identical when $\sigma > -1.4$, the evolution routes are not always the same. To show this, in the next part, we mainly focus on the evolution routes to period-3 and chaos-3 with different initial values.

B. Initial values affect routes

Six initial conditions are chosen to be the same as those in Figs. 7(a)–7(f). The evolution routes to period-3 orbit are given in Figs. 8(a)–8(f) where the parameter $\sigma = -1.8$, and the evolution routes to chaos-3 are presented in Figs. 9(a)–9(f) when the parameter $\sigma = -1$.

For the evolution routes to period-3, Fig. 8 shows that (1) Fig. 8(a) is the same figure as Fig. 3(a), which possesses Route 1: *an initial point* $\rightarrow A_1 \rightarrow A_2 \rightarrow T_c \rightarrow P_3$. (2) When different initial points are chosen, the routes to period-3 are different. They are *an initial point*$\rightarrow A_2 \rightarrow T_c \rightarrow P_3$ in Fig. 8(b), *an initial point* $\rightarrow P_3$ in Figs. 8(c)–8(e), *an initial point*$\rightarrow T_c \rightarrow P_3$ in Fig. 8(f), respectively. From the horizontal axes n, we can see that it takes less time to achieve period-3 if the route is simpler.

For the evolution routes to chaos-3, we notice that (1) Fig. 9(a) is the same route as Fig. 4(b), that is, *an initial point* $\xrightarrow{converge} A_1 \rightarrow A_2 \rightarrow T_c \rightarrow A_3 \rightarrow C_3$. (2) Figure 9(b) shows a different route as *an initial point* $\xrightarrow{converge} A_2 \rightarrow T_c \rightarrow A_3 \rightarrow C_3$, where the part $A_1 \rightarrow A_2$ does not exist in comparison with Fig. 9(a). (3) Although Figs. 7(c) and 7(d) are almost the same bifurcation diagrams, two routes are different in Figs. 9(c) and 9(d) when $y_0 = 2.7$ and $y_0 = 2.75$. The route in Fig. 9(c) is *an initial point* $\xrightarrow{converge} A_3 \rightarrow C_3$, while the route in Fig. 9(d) is *an initial point* $\xrightarrow{converge} A_6 \rightarrow C_3$. (4) Similarly, although Figs. 7(e) and 7(f) have the same bifurcations, two routes are different in Figs. 9(e) and 9(f) when $y_0 = 2.8$ and $y_0 = 3.0$, respectively. The route in Fig. 9(e) is *an initial point* $\xrightarrow{converge} A_6 \rightarrow C_3$, while the route in Fig. 9(f) is *an initial point* $\rightarrow C_3$. Similarly, the less complex the path, the shorter the time to reach chaos-3.

Fourteen evolution routes to period-8 orbit are given in Figs. 17(a)–17(n) in Appendix as $\sigma = 1$. There are some more numerical evolution routes not listed in this paper. From the numerical simulations in this part, we can see that the routes to the period-k or the chaos-k orbit depend on the choice of the initial values. The routes could be parts of Route 1 or 2. Hence, we characterize the intermittent evolution routes, which are different from Route 1 and Route 2 as the following:

Route 3. Possible routes to period-k ($k \geq 3$) orbit in the Rulkov map include *an initial point* $\xrightarrow{converge} A_h$ or $T_c \rightarrow \cdots \rightarrow P_k$, where $h = 1, 2, \ldots, k$ or $2k$. Possible routes to chaos-k ($k \geq 2$) orbit include *an initial point* $\xrightarrow{converge} A_h$ or $T_c \rightarrow \cdots \rightarrow C_k$, where $h = 1, 2, \ldots, k$ or $2k$.

These results in this section illustrate that the asymptotic behaviors of the Rulkov map depend on both the parameters and the initial conditions, particularly the initial values of the slow variable y_0. Different initial values lead to different intermittent evolution routes for the same asymptotic regimes.

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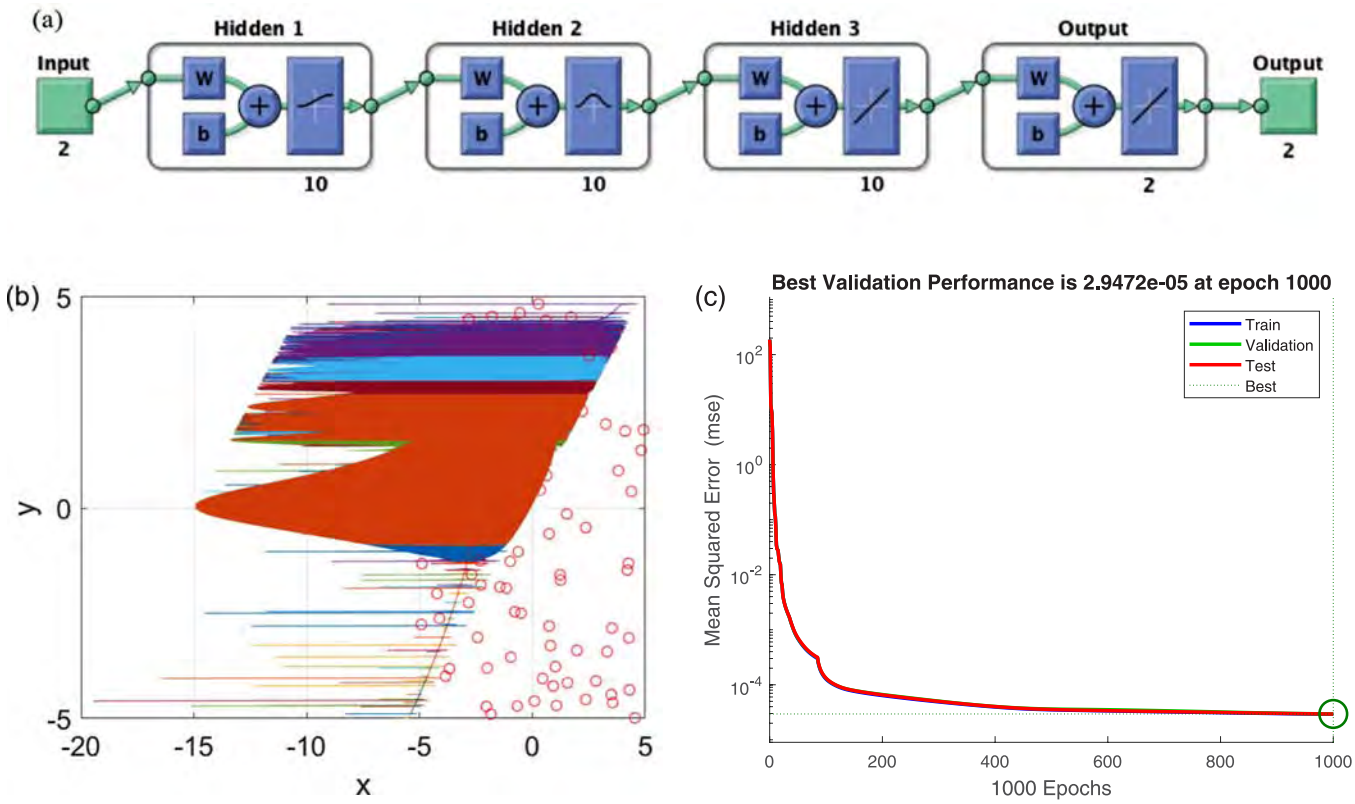

FIG. 10. (a) Network architecture used to train the NN on the trajectory data generated by the Rulkov map. A three-layer network is constructed with ten nodes in each layer and a different activation function for each layer. (b) One hundred trajectories generated from 100 random initial points (the red cycles) when $\alpha = -15$, $\eta = 0.001$, and $\sigma = 1.25$. (c) The performance of the training over 1000 epochs of training.

IV. RECONSTRUCT EVOLUTION ROUTES AND BIFURCATIONS BY MACHINE LEARNING

Neural networks (NNs) offer the abilities to classify and predict the data. Future state predictions by NNs can also be applied to dynamical systems.[30] Feedforward neural network (FFNN) is one of the simplest artificial neural networks. FFNNs connect the input layer to the output layer by forming connections between the units so that they do not form a cycle, while reservoir computing [also known as the echo state networks (ESNs)] is an artificial recurrent neural network with a sparsely connected hidden layer, where each node can feed back on itself.[30,31] Reservoir computing has demonstrated success in time series prediction, inferring unmeasured variables of a dynamical system from measured variable, reconstructing attractor and bifurcation diagrams, etc.[31–35] In this section, we give some experiments to understand the performances of feedforward neural networks and reservoir computing when reconstructing the intermittent evolution routes and the bifurcations of the Rulkov map.

A. Feedforward neural networks

The state of the Rulkov map is given by $\mathbf{x} = [x, y]^T$. The goal of this part is to train a neural network to learn an evolution rule which maps the state space from $\mathbf{x}_k$ to $\mathbf{x}_{k+1}$, where k is the discrete time. The training data required for the NNs are constructed from the time series produced by the Rulkov map using numerical simulation. The parameters of the Rulkov map are fixed at $\alpha = -15$, $\sigma = -1.8$, and $\eta = 0.001$, which will produce intermittent evolution routes to period-3 or period-8 orbit depending on the initial values. In other words, the input matrix is $\mathbf{x}_k$, while the output matrix is the corresponding state $\mathbf{x}_{k+1}$ in the intermittent evolution routes to asymptotic orbits. One hundred trajectories are generated from 100 initial points, which are randomly chosen in the region $(x_0, y_0) \in [-5, 5] \times [-5, 5]$ [the red cycles in Fig. 10(b)]. Each trajectory is composed of 10 001 points as time $n = 0, \ldots, 10\,000$, since 10 001 points are enough to reveal the whole routes to period-3 or period-8. Therefore, we have $100 \times 10\,000$ pairs of input matrices $\mathbf{x}_k$ and output matrices $\mathbf{x}_{k+1}$ $(k = 0, 1, \ldots, 9999)$ to train the NN.

A three-layer feedforward neural network, as shown in Fig. 10(a), is constructed with ten nodes in each layer, whose activation units are differently chosen as logsig, radbas, and purelin functions. The number of epochs is set to 1000 and the maximum failure times is six in Neural Network Toolbox in MATLAB. The performance of the training is drawn in Fig. 10(c). A function *net* is then produced, which is used to predict the future state of the Rulkov map.

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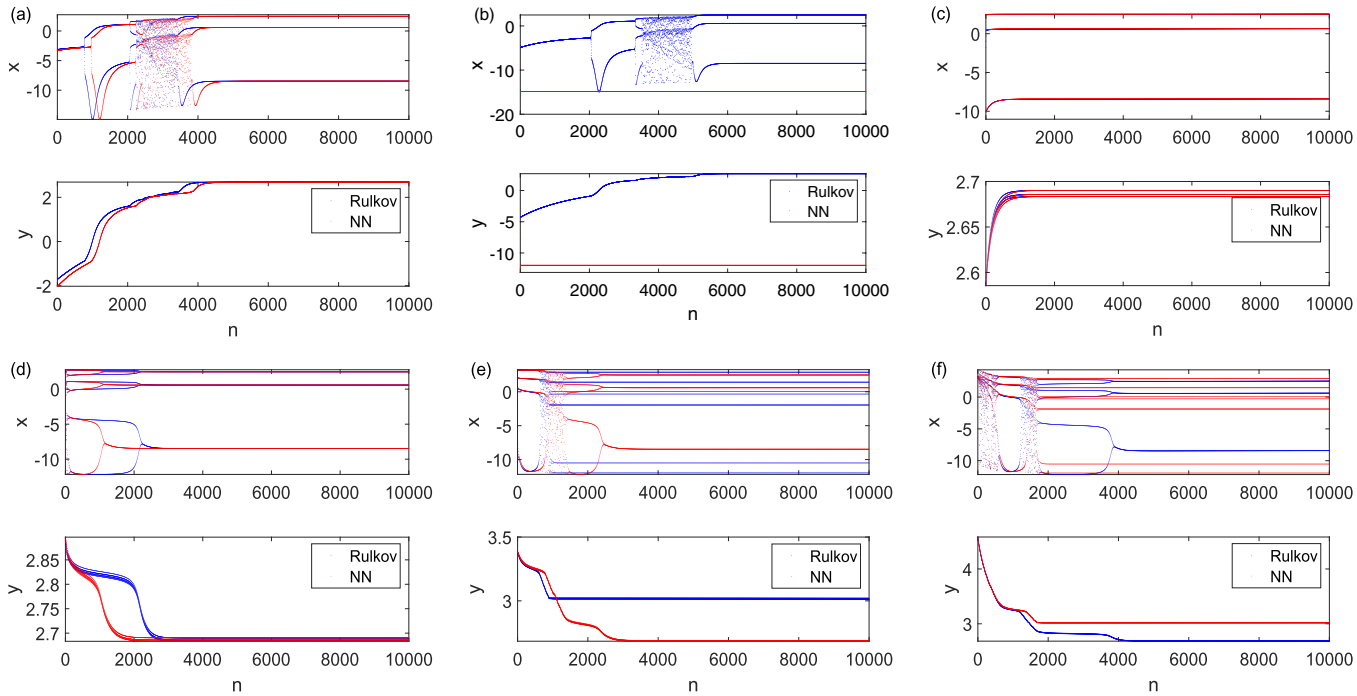

FIG. 11. (a)–(f) The trajectories generated by the numerical simulation against the feedforward neural network prediction from six randomly chosen initial conditions.

Thirty initial values are randomly taken in the region $(x_0, y_0) \in [-5, 5] \times [-5, 5]$ to test the accuracy of the FFNN prediction of the trajectories. Six of them are shown in Figs. 11(a)–11(f). We can see three kinds of differences between the trajectories generated by the numerical simulation and the trajectories predicted by FFNNs: (1) Some of them, like those in Figs. 11(a), 11(c), and 11(d), successfully predict both the evolution routes and the asymptotic orbits, except for a few time differences. (2) Some of them predict parts of routes but present different asymptotic orbits, such as those in Figs. 11(e) and 11(f), where the period-8 orbits are falsely predicted to be convergent to period-4. (3) Some of them, like that in Fig. 11(b), are wrongly predicted all the time. The other 24 results have almost the same cases as these 6 cases and hence we do not show all of them in this paper. Only 17 of the trajectories can be learned the types of both routes and the asymptotic orbits. We may say that the accuracy rate is about 60%. It is worthwhile to point out that the mean squared errors (MSEs), the accuracy rates, and the time required for training are not all the same when using different versions of MATLAB or different computers.

To improve the performance of NNs, it is trivial to make the network deeper and wider and add the number of training data and epochs.[30] However, the weakness of FFNNs of the above experiments are as the following. First, it takes more than an hour to train the neural network when considering 100 trajectories over 1000 epochs using a desktop computer with 64-bit Windows system and 16G of memory. It will be more time-consuming to learn more complex systems. Second, these experiments also remind us of a disadvantage of machine learning, that is, only the trajectories in finite time can be trained. Third, the initial conditions are arbitrary, such that one may not train the sufficient number of trajectories to guarantee accurate prediction on the routes, especially for high dimensional systems.

Therefore, we will use the reservoir computing to reconstruct the evolution routes and the bifurcations of the Rulkov map, the spatiotemporal patterns of a network coupled by Rulkov maps in the next part.

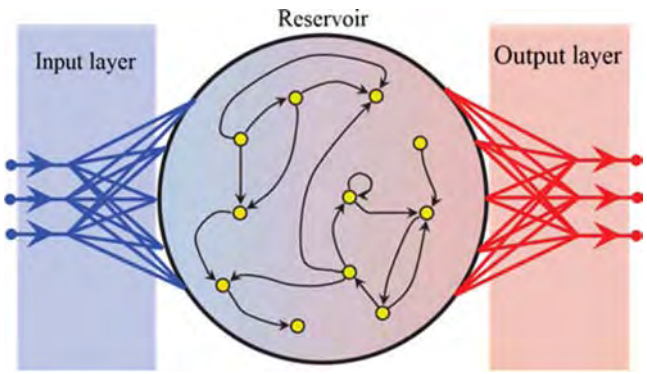

FIG. 12. The architecture of a reservoir computing.[33]

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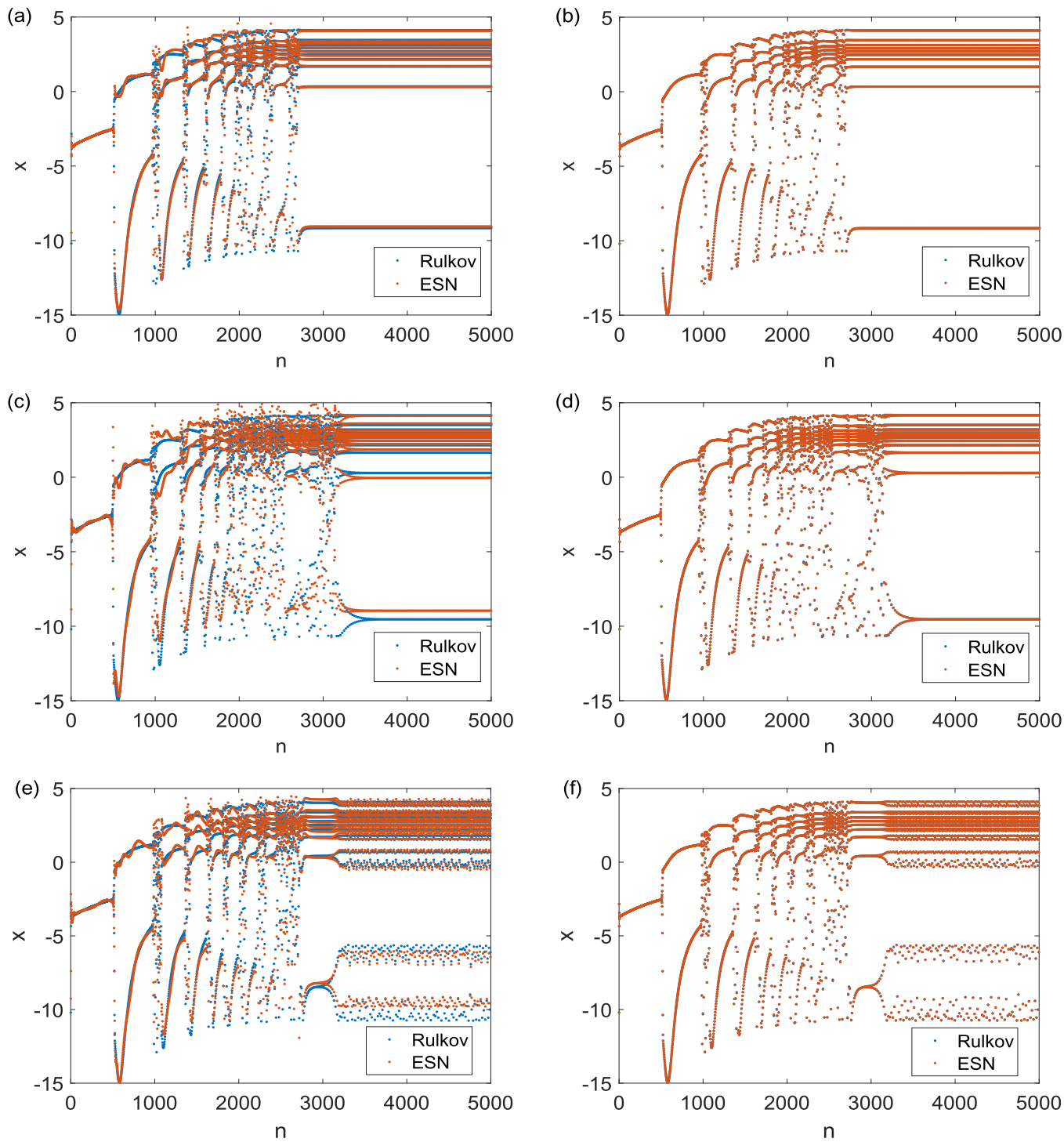

FIG. 13. Evolution routes from $x_0 = 0.1$, $y_0 = -2.7$ to (a) period-10 orbit as $\sigma = 1.38$, $N_r = 50$; (b) period-10 orbit as $\sigma = 1.38$, $N_r = 500$; (c) period-11 orbit as $\sigma = 1.48$, $N_r = 50$; (d) period-11 orbit as $\sigma = 1.48$, $N_r = 500$; (e) chaos-9 orbit as $\sigma = 1.30$, $N_r = 50$; and (f) chaos-9 orbit as $\sigma = 1.30$, $N_r = 500$.

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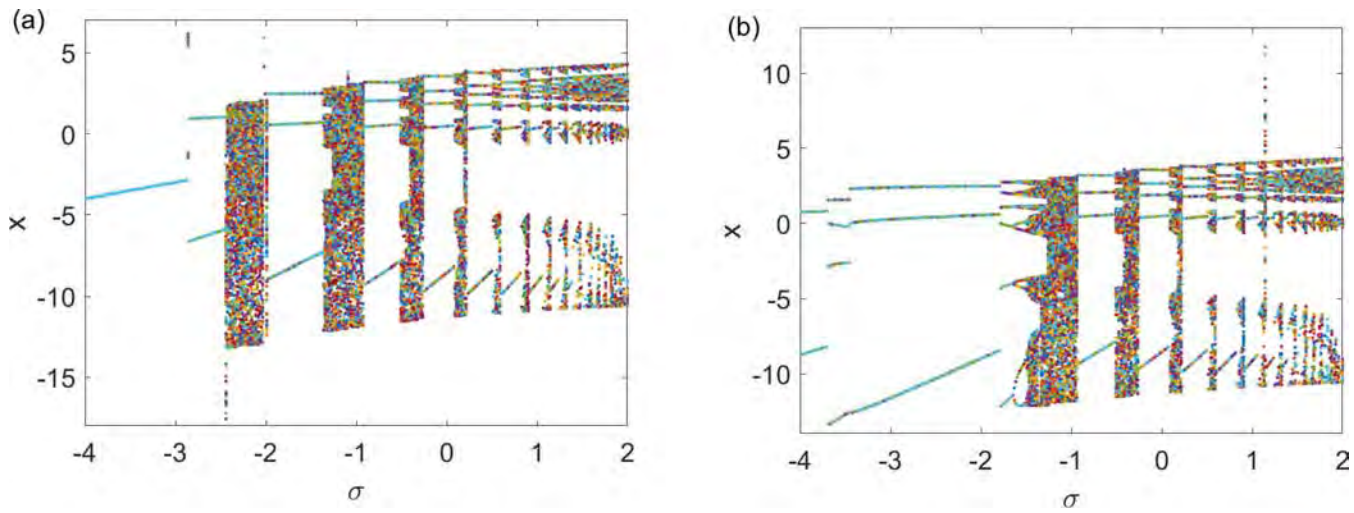

FIG. 14. Bifurcation diagrams reconstructed by reservoir computing when $\alpha = -15$, (a) $x_0 = 0.1$, $y_0 = -2.7$ and (b) $x_0 = -1$, $y_0 = 3.0$.

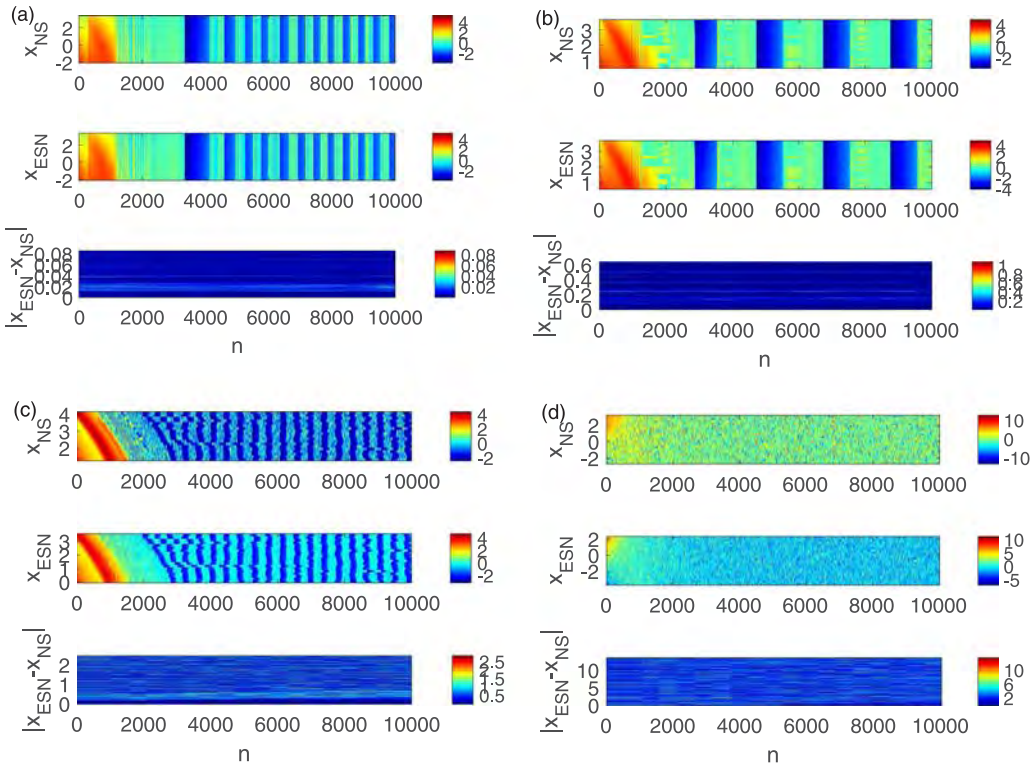

FIG. 15. Spatiotemporal patterns generated by numerical simulation, reservoir computing, and the errors between them in the globally coupled network. (a) Complete synchronization as the coupling strength $\varepsilon = 0.03$. (b) In-phase synchronization as $\varepsilon = 0.02$. (c) Nearly in-phase synchronization as $\varepsilon = 0.001$. (d) All neurons are in chaotic non-synchronous spiking states as $\varepsilon = 0.2$ for $n = 0, \ldots, 10\,000$.

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B. Reservoir computing

The architecture of reservoir computing is shown in Fig. 12. In this subsection, we will explore the performances of reservoir computing when a single Rulkov neuron model and a Rulkov neuron network model are considered, respectively. The features of the ESNs include (1) offline training with pseudo-inverse or ridge regression; (2) multiple time series supported; (3) two different parameters for input and bias scaling; and (4) control of reservoir density. The properties of ESNs with leaky integrator neurons are as follows: the leakage rate is 0.2 and the spectral radius is 0.9. The original MATLAB codes we cited can be found in GitHub.[36]

The trajectories composed of 10 001 points are generated from a fixed initial point as time $n = 0, \ldots, 10\,000$. The input of ESNs is $\mathbf{x}_k$ $(k = 0, 1, \ldots, 9999)$, while the target is the corresponding state $\mathbf{x}_{k+1}$. Figures 13(a)–13(f) show the comparisons of the routes from initial values $x_0 = 0.1$ and $y_0 = -2.7$ to period-10, period-11, and chaos-9 orbits of the Rulkov map when $N_r = 50$ and $N_r = 500$, respectively. One can see that a smaller size of reservoir computing leads to obvious differences between the blue and the red waveforms in Figs.13(a), 13(c), and 13(e). However, a larger size leads to smaller differences in Figs. 13(b), 13(d), and 13(f), i.e., a larger size of reservoir computing N_r can reduce the differences between the numerical and the predicted evolution routes but needs more time. It takes about half a second to obtain Figs. 13(a), 13(c), or 13(e) as $N_r = 50$, while about one second to obtain Figs. 13(b), 13(d), or 13(f) as $N_r = 500$.

Figure 14 shows the bifurcation diagrams reconstructed by reservoir computing when the initial values are (a) $x_0 = 0.1, y_0 = -2.7$ and (b) $x_0 = -1, y_0 = 3$, respectively. For the Rulkov model, reservoir computing almost predicts the exact asymptotic dynamical behaviors. Period-adding bifurcation is successfully reconstructed by reservoir computing. Compared with the bifurcation diagrams generated by numerical calculation, some extra points are predicted when the parameter σ is fixed at some of the bifurcation values. One possible reason is that the matrices in the ridge regression file could become singular as the codes are invoked. This warning can be displayed in the command window of MATLAB. It takes about 6 min to obtain each figure in Fig. 13.

Finally, we report the effects on the errors of a higher dimension discrete-time system based on the Rulkov neuron network model. As many neuron network models, coupled Rulkov networks present complete synchronization, in-phase synchronization, nearly in-phase synchronization or chaotic non-synchronous spiking state, etc.[37–39] We consider a Rulkov neuron network, which is globally coupled by 19 neurons.[39] Figures 15(a)–15(d) show the spatiotemporal patterns of complete synchronization, in-phase synchronization, nearly in-phase synchronization and chaotic non-synchronous spiking states of the coupled network, respectively. Comparing the differences of transmembrane voltage $|x_{ESN} - x_{NS}|$, where x_{ESN} presents the trajectories predicted by reservoir computing and x_{NS} presents the trajectories generated by numerical simulation, we find that the maximal difference of transmembrane voltage $|x_{ESN} - x_{NS}|$ increases as the complete synchronous network becomes in-phase synchronization, nearly in-phase synchronization, and chaotic non-synchronous spiking states. Non-synchronous networks have higher complexity; thus, the synchronous networks can be predicted more accurately than the non-synchronous networks.

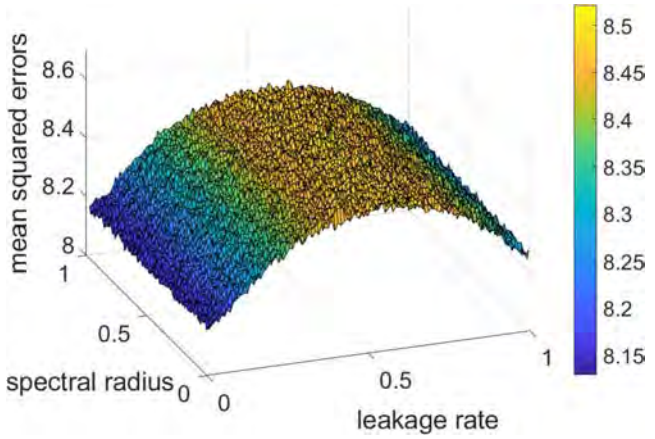

FIG. 16. Mean squared errors $\sum_{i=1}^{19} MSE_{x_i}$ when the leakage rate and the spectral radius are considered as varying parameters.

Figure 16 shows the mean squared errors $\sum_{i=1}^{19} MSE_{x_i}$ when the leakage rate and the spectral radius of the reservoir computing are considered as varying parameters. The spectral radius has little effect on MSEs, while a leakage rate near 0.5 allows the MSE to reach a maximal value. Nevertheless, the MSE of 19 neuron network has small changes as the parameters of reservoir computing varies.

V. CONCLUSIONS

In this paper, we report the intermittent evolution routes from initial conditions to the asymptotic regimes in the Rulkov map. The windows with transient approximate periodic and transient chaotic behaviors occur before the system reaches the periodic or the chaotic orbits. The routes from an initial condition to a period-k $(k \geq 3)$ or a chaotic orbit are summarized as Route 1 and Route 2, of which the evolution routes to chaotic orbits can be classified according to the windows before reaching asymptotic chaotic regimes. Furthermore, Route 3 might be a part of the whole routes presented in Route 1 and Route 2. Since initial values affect the asymptotic behaviors and the evolution routes of the system, one may not accurately predict the routes with random initial conditions by feedforward neural networks. However, if the initial values are given, then the routes to the periodic or the chaotic orbits, the period-adding bifurcation, and the synchronous network can be reconstructed by reservoir computing. A nonsingular parameter condition, a larger size of reservoir computing, or a simpler orbit for the system will lead to less difference between the trajectories generated by numerical simulation and predicted by reservoir computing.

One can develop some meaningful research studies based on these results in this paper: (1) whether there are other discrete-time systems with the same evolution routes to the periodic or the chaotic

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orbits, especially, whether the Poincaré maps of other continuous-time neuron models have the same evolution routes to the periodic or the chaotic orbits. (2) If we want to control a discrete-time system so that it admits period-k ($k \geq 3$) or chaos-k ($k \geq 2$) orbit, we can add the controller in a stage that is close to the end of the routes *an initial point* $\xrightarrow{converge} A_1 \to A_2 \to T_c \to A_3 \to T_c \to \cdots \to T_c \to A_{k-1} \to T_c \to$ *period-k (or* $\ldots \to A_k \to$ *Chaos-k)*, for example, adding the controller at the stage A_{k-1}. It may take less time to achieve a period-k or a chaos-k orbit than adding the controller at the stage A_1. (3) We have classified the routes to chaotic orbits according to the windows before reaching asymptotic chaotic regimes. However, is there any difference among chaos-k ($k \geq 2$) if k is fixed at different integers? Accordingly, what do they correspond to in the real world? Further research may help us to understand the trigger of neuron firing, abnormal synchronization, even to control them better.

ACKNOWLEDGMENTS

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The authors have no conflicts to disclose.

APPENDIX: EVOLUTION ROUTES TO PERIOD-8 ORBIT

Evolution routes to period-8.

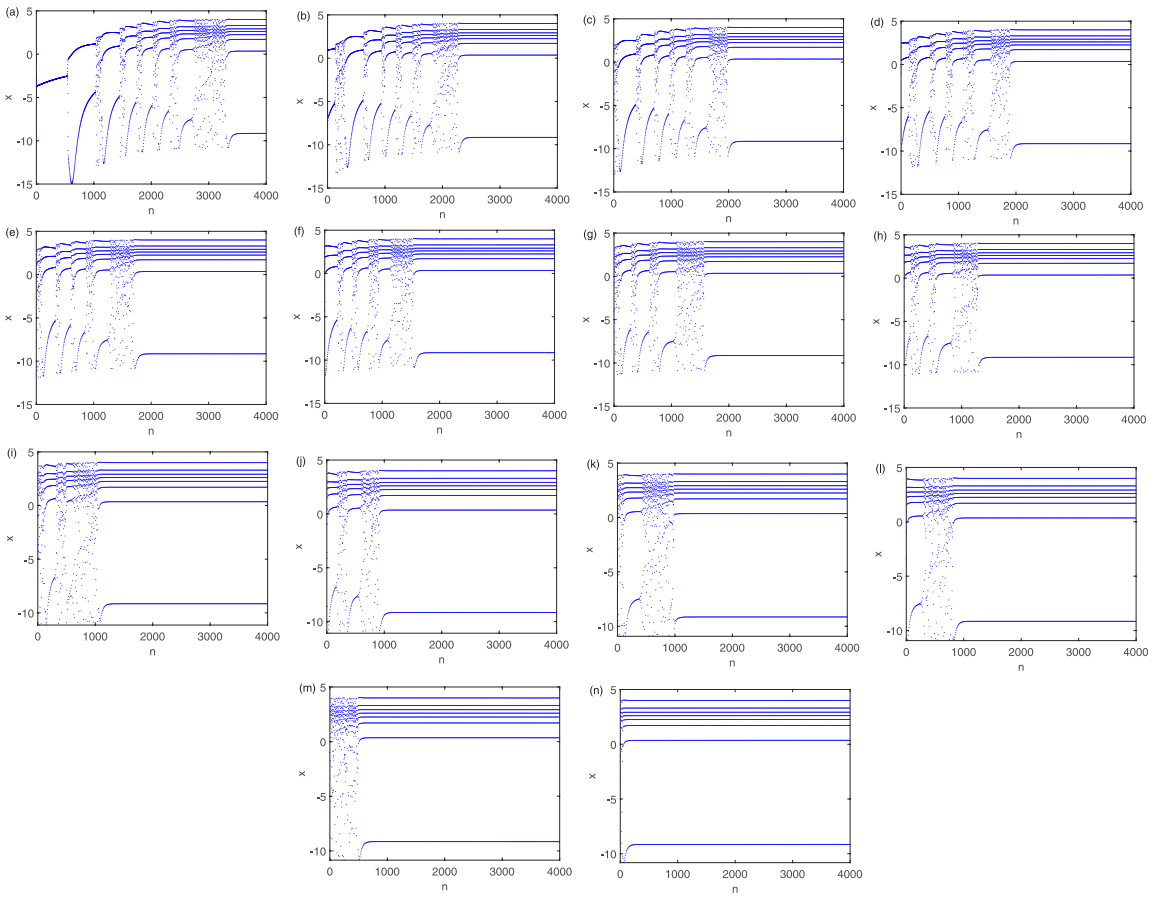

FIG. 17. Evolution routes to period-8: (a) *an initial point* $\to A_1 \to A_2 \to T_c \to A_3 \to T_c \to A_4 \to T_c \to A_5 \to T_c \to A_6 \to T_c \to A_7 \to T_c \to$ *period-8* when $y_0 = -2.7$; (b) *an initial point* $\to A_2 \to T_c \to A_3 \to T_c \to A_4 \to T_c \to A_5 \to T_c \to A_6 \to T_c \to A_7 \to T_c \to$ *period-8* when $y_0 = 1.2$; (c) *an initial point* $\to T_c \to A_3 \to T_c \to A_4 \to T_c \to A_5 \to T_c \to A_6 \to T_c \to A_7 \to T_c \to$ *period-8* when $y_0 = 2.0$; (d) *an initial point* $\to A_3 \to T_c \to A_4 \to T_c \to A_5 \to T_c \to A_6 \to T_c \to A_7 \to T_c \to$ *period-8* when $y_0 = 2.6$; (e) *an initial point* $\to T_c \to A_4 \to T_c \to A_5 \to T_c \to A_6 \to T_c \to A_7 \to T_c \to$ *period-8* when $y_0 = 3.0$; (f) *an initial point* $\to A_4 \to T_c \to A_5 \to T_c \to A_6 \to T_c \to A_7 \to T_c \to$ *period-8* when $y_0 = 3.25$; (g) *an initial point* $\to T_c \to A_5 \to T_c \to A_6 \to T_c \to A_7 \to T_c \to$ *period-8* when $y_0 = 3.5$; (h) *an initial point* $\to A_5 \to T_c \to A_6 \to T_c \to A_7 \to T_c \to$ *period-8* when $y_0 = 3.7$; (i) *an initial point* $\to T_c \to A_6 \to T_c \to A_7 \to T_c \to$ *period-8* when $y_0 = 3.8$; (j) *an initial point* $\to A_6 \to T_c \to A_7 \to T_c \to$ *period-8* when $y_0 = 3.91$; (k) *an initial point* $\to T_c \to A_7 \to T_c \to$ *period-8* when $y_0 = 4.0$; (l) *an initial point* $\to A_7 \to T_c \to \to$ *period-8* when $y_0 = 4.06$; (m) *an initial point* $\to T_c \to$ *period-8* when $y_0 = 4.1$; and (n) *an initial point* $\to$ *period-8* when $y_0 = 4.15$. The other parameters and initial values are the same: $\sigma = 1.0$ and $x_0 = -1$.

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DATA AVAILABILITY

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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